

Volume II · Chapter 8 — Prompt Engineering for Medicine · Labs

Source: split from med-tech-curriculum-V1.md (V1). From this split onward this chapter is maintained **independently** here. See Volume-II-Chapter-8-context.md for the AI interaction log.

Prerequisites: Ch 7.

Learning outcomes — the student can: design effective prompts (zero/few-shot, chain-of-thought); produce structured outputs; assess reliability and hallucination; apply basic safety/injection awareness in clinical apps.

Labs (hands-on) — 10 h:

#	Lab	h
8a	Prompt iteration on a clinical QA task; zero-vs. few-shot	3
8b	Chain-of-thought vs. direct on clinical reasoning; measure	2
8c	Structured extraction: prompt an LLM to output a JSON clinical summary	3
8d	Hallucination probe: design tests, log failures	2

Datasets/tools: local LLM via Ollama (MedGemma) or HF; Python. **Assessment:** prompt-portfolio deliverable (**50%**); structured-extraction lab (**30%**); quiz (**20%**). **Key**

decisions: prompting vs. fine-tuning vs. RAG; how to measure reliability; guardrails.

References: plan §13 → *Medical LLMs*. **Hours:** Theory 8 + Lab 10 = 18.